

UNIVERSITY OF MILANO-BICOCCA

Numerical Simulations of the Ising model through Monte Carlo Markov Chains



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Abstract

This work presents the numerical study of the ferromagnetic Ising model in one, two, and three dimensions to observe phase transitions as a function of the lattice size. The system configurations were obtained by sampling a Markov Chain at fixed temperature generated with the Metropolis-Hastings algorithm following an analysis of its thermalisation and autocorrelation time. The critical temperature was estimated in the two and three dimensional cases by analyzing the finite size scaling of the magnetization. Analogously, the absence of a finite temperature phase transition was found in the one dimensional case, in accordance with analytical results.

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A brief history of the Ising model

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Chapter 1

Theoretical background

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Chapter 2

Methods

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Chapter 3

Results

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Appendix A

Appendix

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Bibliography

[1] Author, Title, Journal/Book, Year.